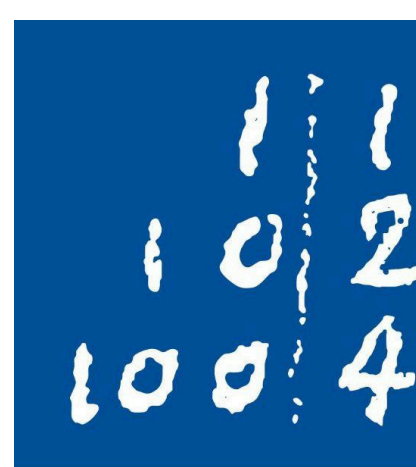


When and Why Does Hindsight Experience Replay Improve Soft Actor-Critic under Sparse Rewards?

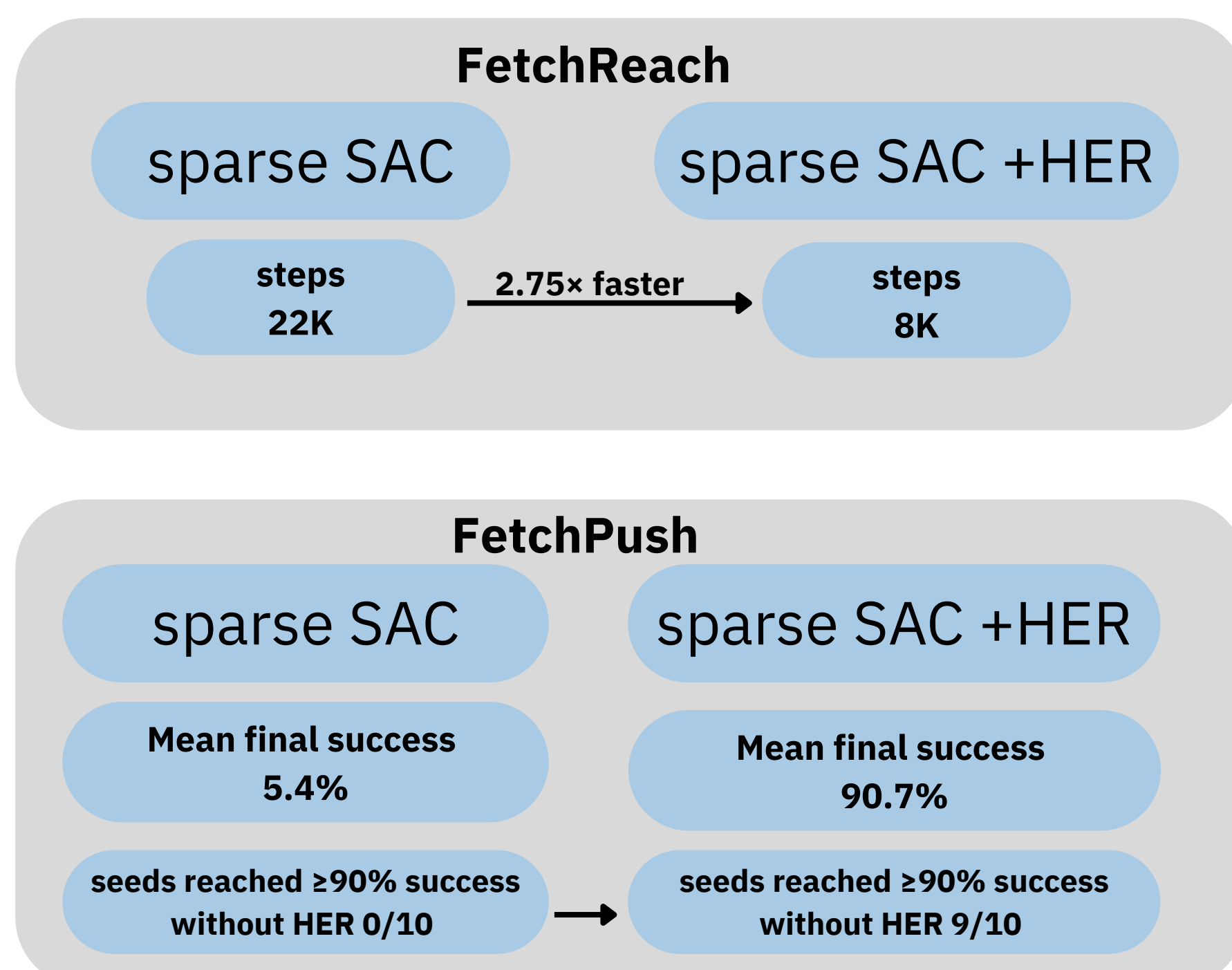


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Poster Presentations in context of ADRL Lecture

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TL;DR



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Motivation & Problem Setting

Motivation

- Sparse rewards provide very little learning signal, which makes successful behavior hard to discover. [Andrychowicz et al., HER, 2017]
- SAC can spend many steps learning from failed experiences.
- HER reuses failed experiences by changing the goal to one that was actually reached.

Problem Setting

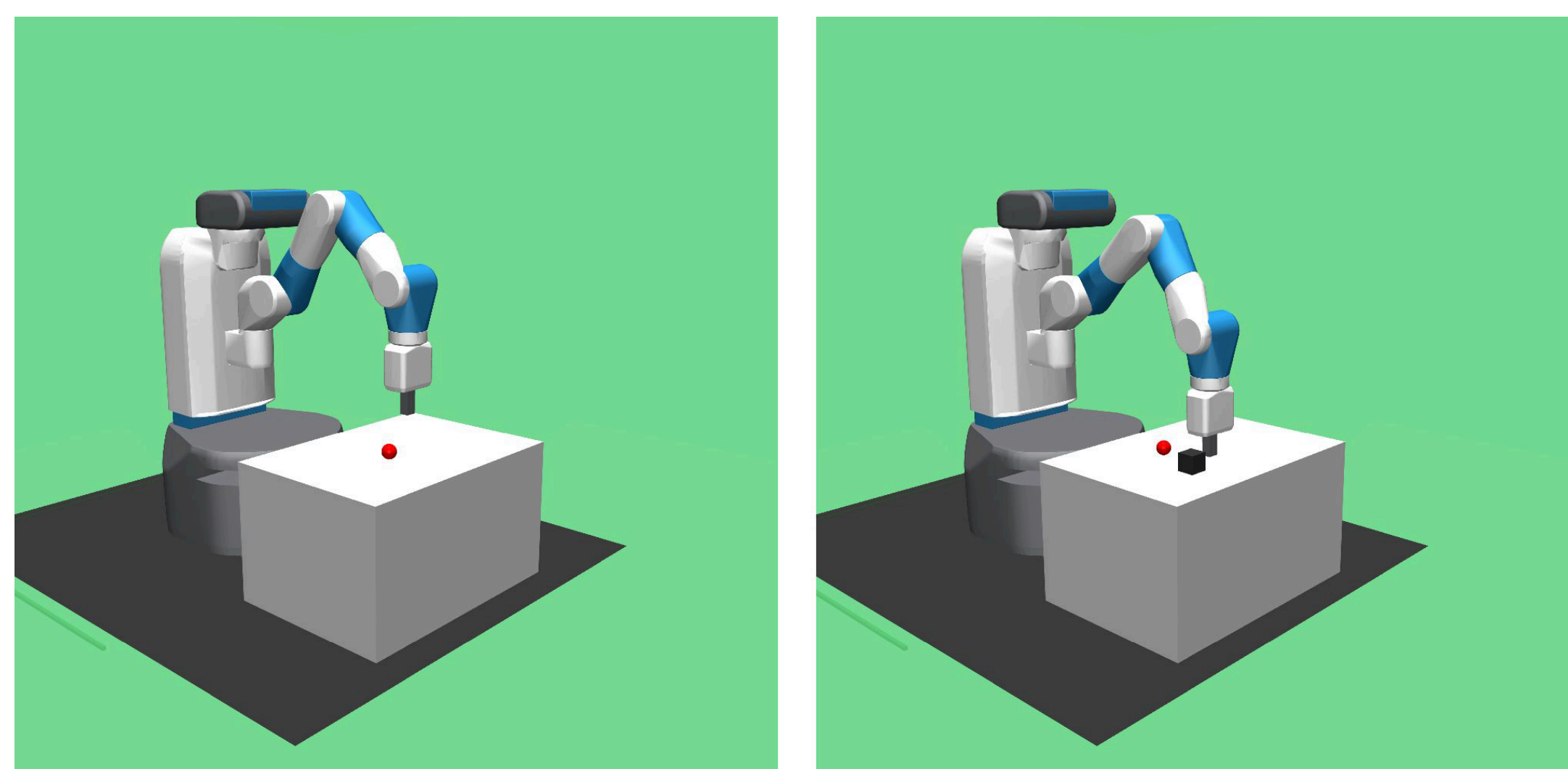
- Compare easy FetchReach-v3 vs. harder FetchPush-v3.
- Compare sparse SAC vs. sparse SAC + HER.
- Ask:
 - When does HER matter most?
 - Why does it help?
 - Does more relabeling always help?

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Approach

- FetchReach-v3 (easy): 4 conditions [dense], [dense+HER], [sparse], [sparse+HER].
- FetchPush-v3 (hard): 2 conditions [sparse], [sparse+HER].
- SAC (Stable-Baselines3) + HerReplayBuffer, using the same hyperparameters across conditions.
- 70 main runs total, with 10 seeds per main condition and 5 seeds per ablation setting.
- Track SAC's entropy coefficient α and the goal-distance distribution to study the training dynamics.
- HER ablation:

$n_{\text{sampld_goal}} \in \{1, 4, 8\}$



FetchPush-v3

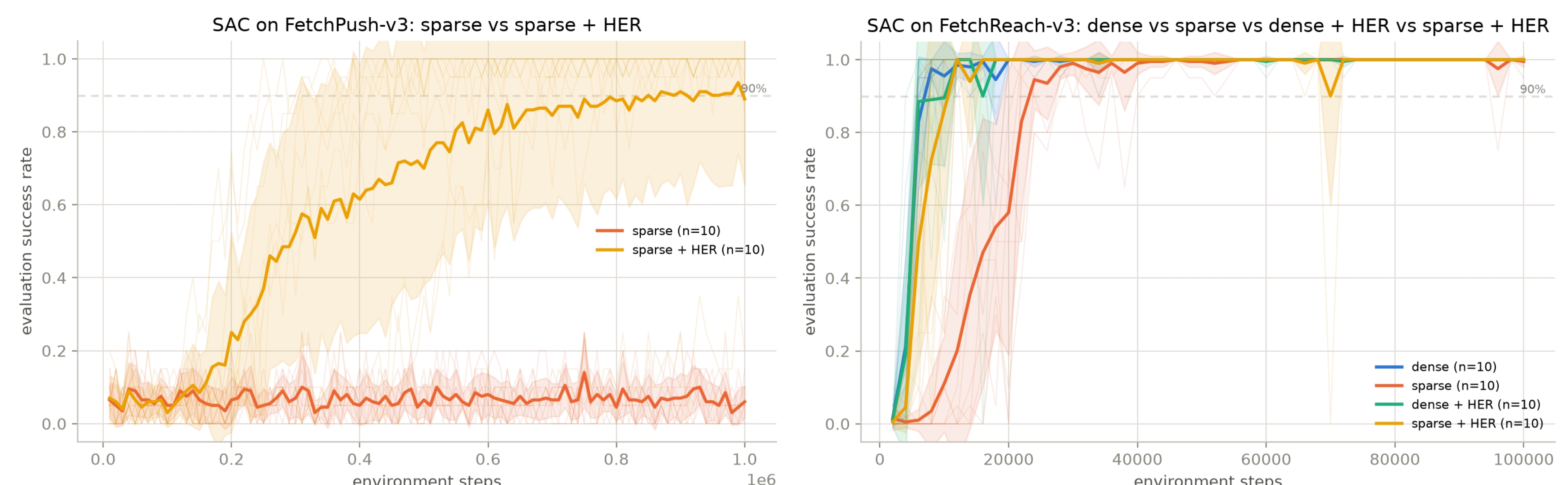
FetchReach-v3

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Key Insights

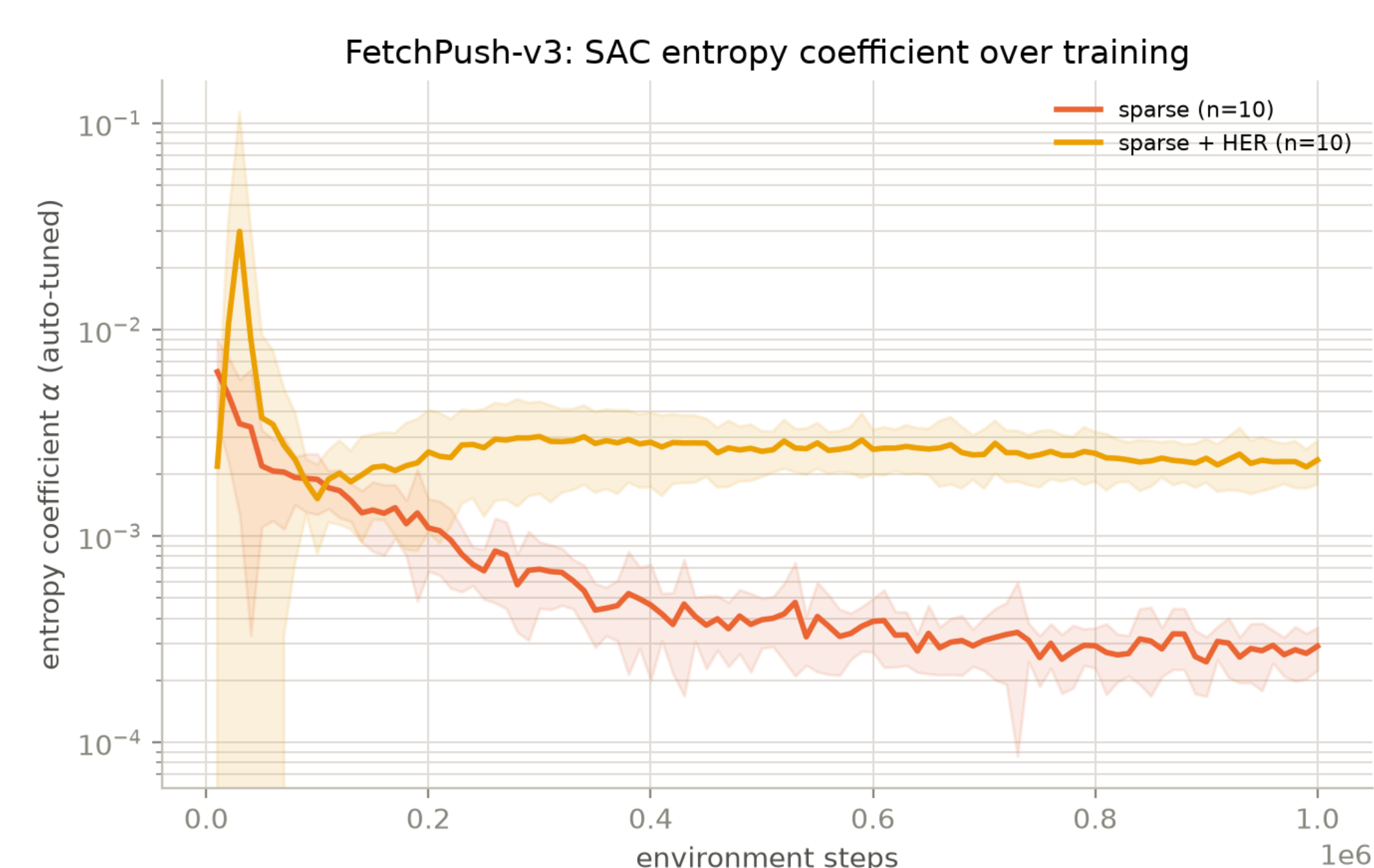
Key Insight 1: HER matters more on the harder task

- HER is a modest speedup on FetchReach, but much more important on FetchPush.



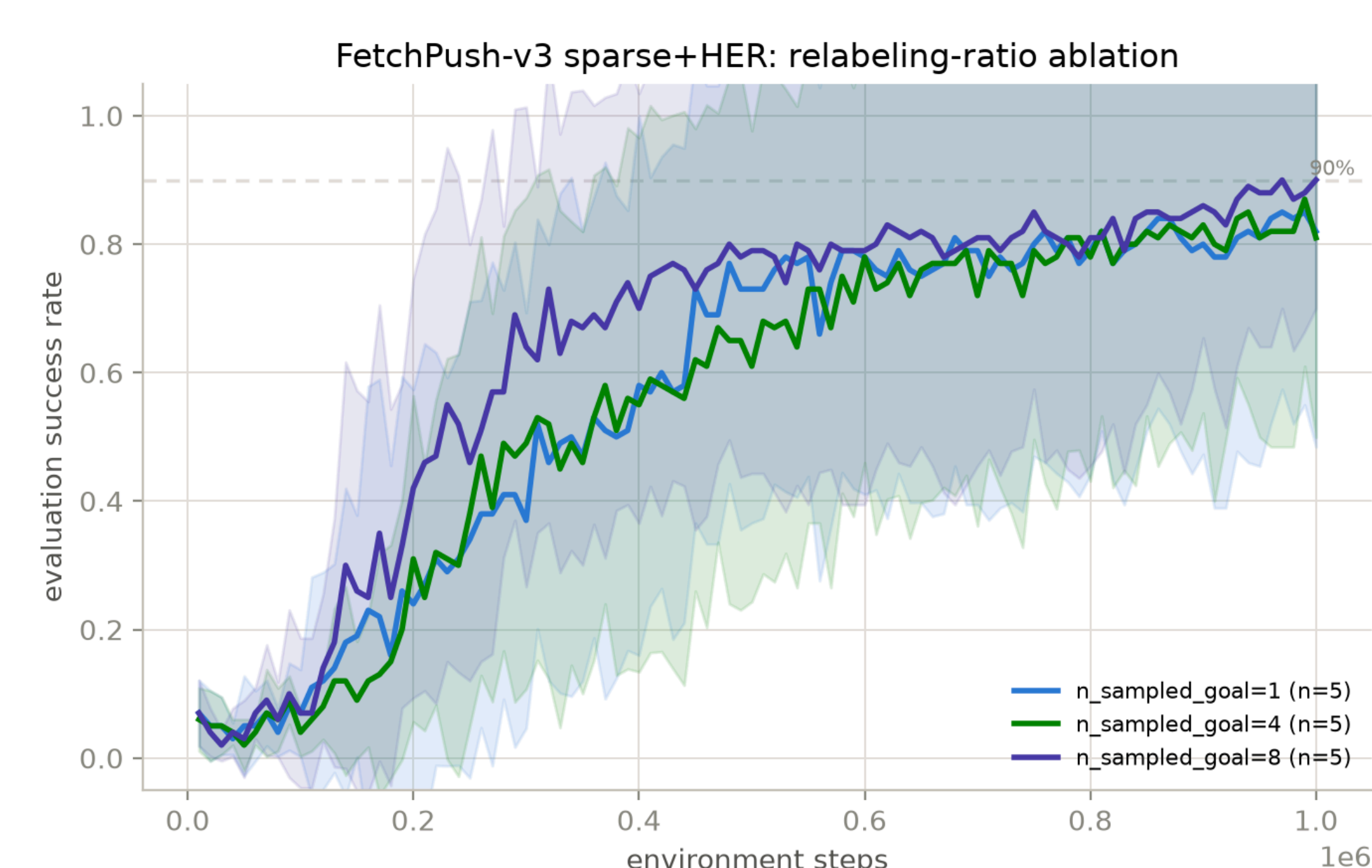
Key Insight 2: HER is linked to healthier entropy dynamics

- On FetchPush, SAC without HER becomes low-entropy while it is still failing.



Key Insight 3: More relabeling is not always better

- Increasing HER relabeling helps mainly with speed, not final performance.



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Future Works

- Cleaner curriculum metric (real-goal-only transitions) for Push
- Test a harder Fetch task
- Statistical significance across seeds



Code, data & full report
github.com/mustafa99705/-adrl-sac-her-fetchreach